SFWRENG 3MX3 2025 Midterm 2 Solutions

Matthew Farah, ⟨farahm11@mcmaster.ca⟩, 400468251

December 6, 2025

Contents

Question 1 - Block Diagrams 2

Question 2 - The Fourier Transform 5

Question 3 - Short Answer Questions 7
- 3.a . . . 7
- 3.b . . . 7
- 3.c . . . 7
- 3.d . . . 7

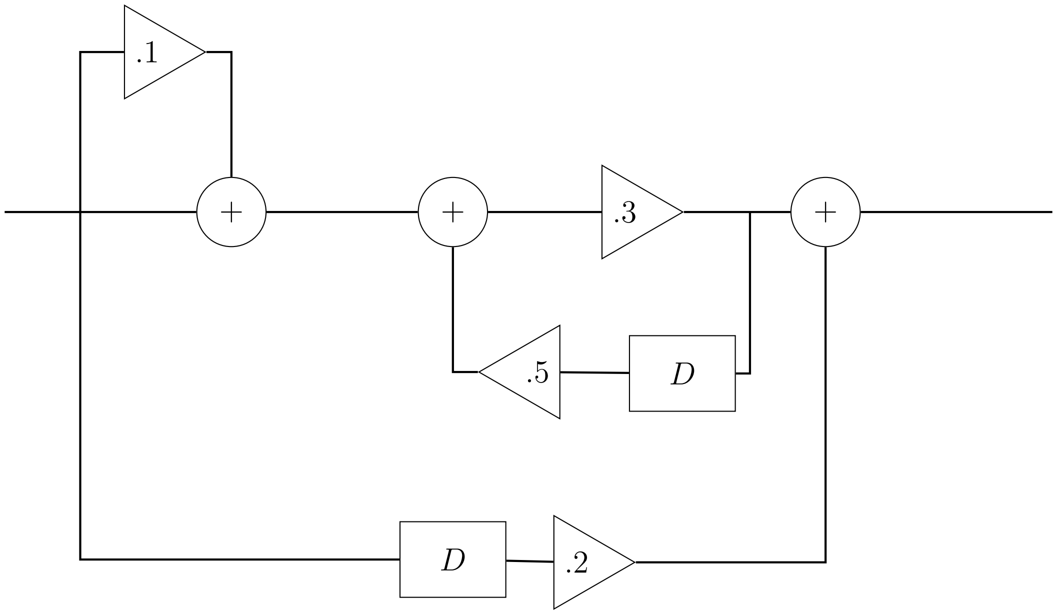
Question 4 - Modulation 9

Question 5 - Filters 11
- 5.a . . . 11
- 5.b . . . 11

Question 6 - The Discrete Fourier Transform 13

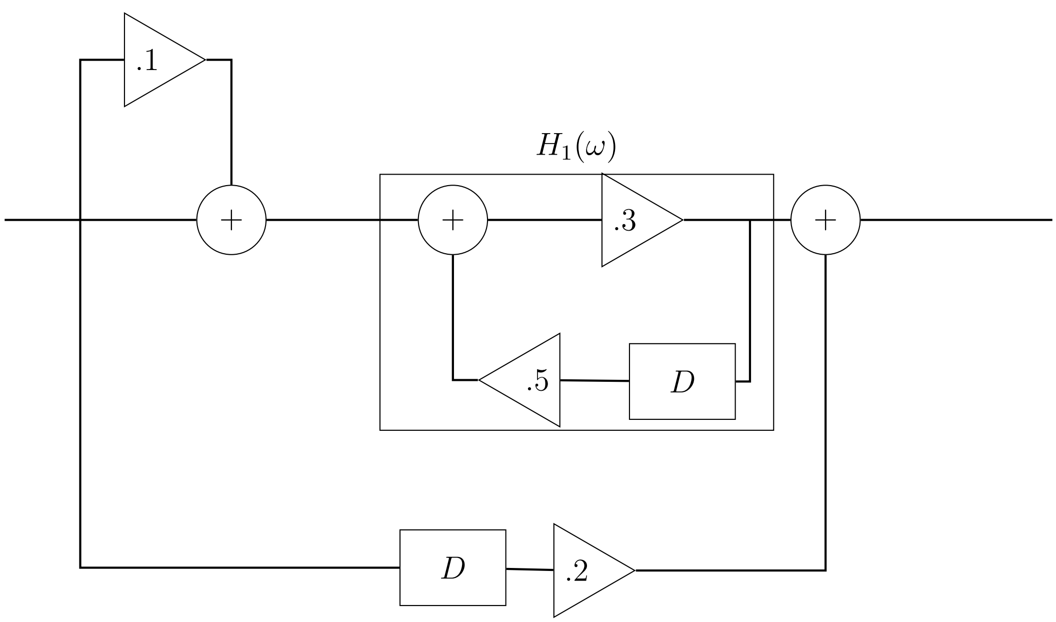
Question 1

Given the following block diagram of a discrete system, find its frequency response.



Solution:

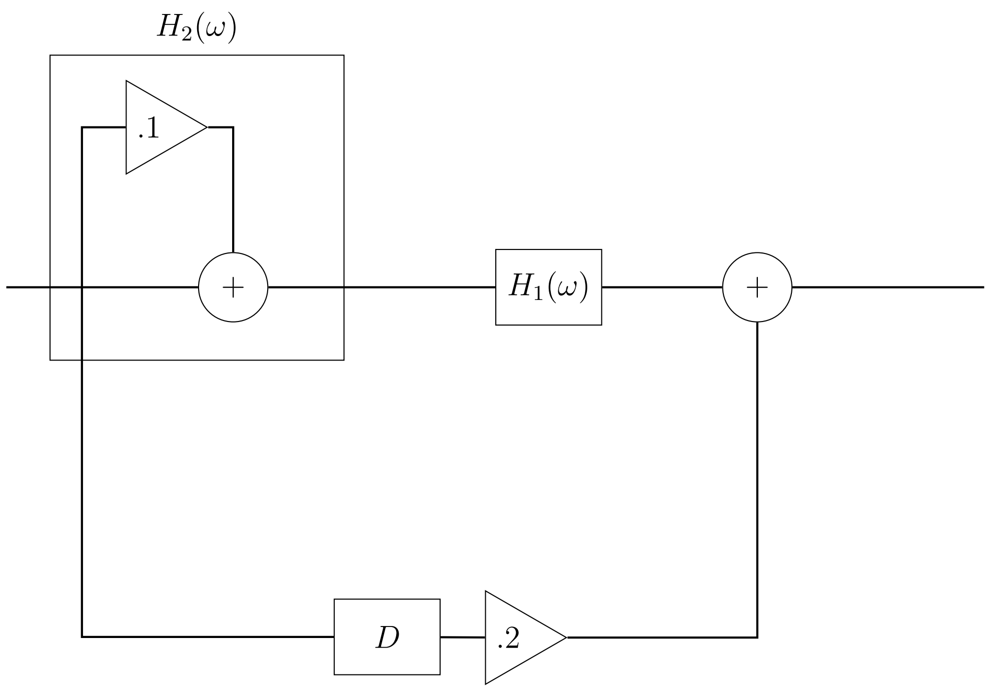
First, we simplify the block diagram by computing the frequency response, $H(\omega)$, of the indicated portion of the block diagram:



Since this is a simple feedback loop, we can determine that:

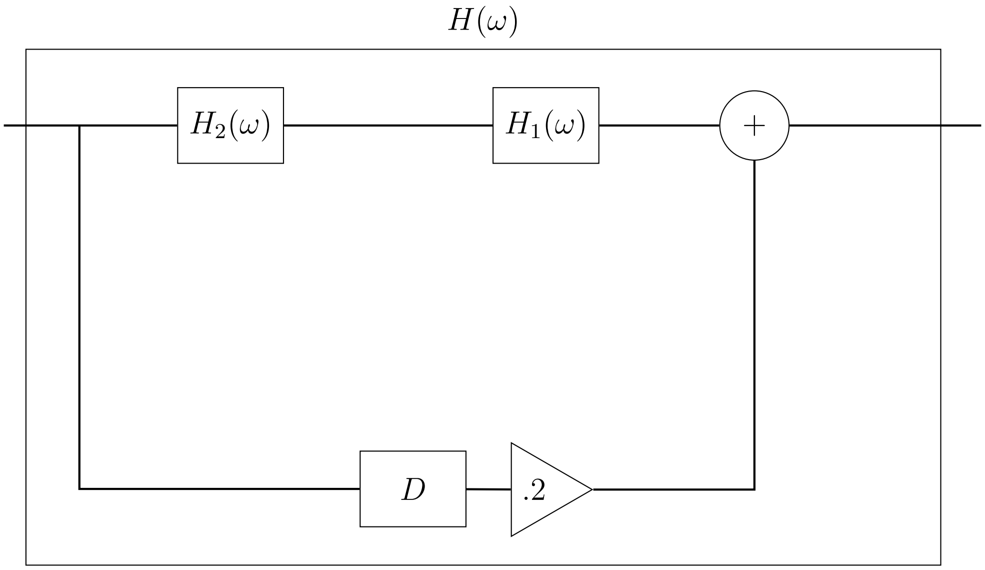
$$\begin{aligned} H_1(\omega) &= \frac{\frac{3}{10}}{1 - \left(\frac{3}{10}\right)\left(\frac{1}{2}\right)e^{-i\omega}} \\ &= \frac{3}{10 - \frac{3}{2}e^{-i\omega}} \end{aligned}$$

Note here that the delay of one sample corresponds to $e^{-i\omega}$.
We can now see that the block diagram has been simplified down to the following, once again, I have indicated the next part of the diagram to simplify:



$$\begin{aligned} H_2(\omega) &= \frac{1}{10} + 1 \\ &= \frac{11}{10} \end{aligned}$$

And we continue on in the same manner as before, giving us the frequency response of the entire system.



$$
\begin{aligned}
H(\omega) &= H_1(\omega)H_2(\omega) + \frac{1}{5}e^{-i\omega} \\
&= \frac{3}{10 - \frac{3}{2}e^{-i\omega}} \cdot \frac{11}{10} + \frac{1}{5}e^{-i\omega} \\
&= \frac{33}{100 - 15e^{-i\omega}} + \frac{1}{5}e^{-i\omega}
\end{aligned}
$$

Question 2

Given is a filter by its impulse response:

$$h(t) = \begin{cases} 1, \text{ if } -1 \leq t \leq 0 \\ -1, \text{ if } 1 \leq t \leq 2 \\ 0, \text{ otherwise} \end{cases}$$

Compute the frequency response, $H(\omega)$ of the system.

Solution:

We know that the frequency response

$$H_0(\omega) = \begin{cases} \frac{1}{2a}, \text{ if } -1 \leq t \leq a \\ 0, \text{ otherwise} \end{cases}$$

corresponds to a signal whose impulse response is given by $\text{sinc}(a\omega)$. Thus, by considering manipulations of $H_0(\omega)$, we can find the frequency response of the system without having to directly compute it.

(I) Find the frequency response for $-1 \leq t \leq 0$.

If we set $a = \frac{1}{2}$, and advance the impulse response by half a second, we would get an impulse response corresponding to $h(t) = \begin{cases} 1, \text{ if } -1 \leq t \leq 0 \\ 0, \text{ otherwise} \end{cases}$, thus giving that:

$$H_2(\omega) = \text{sinc}\left(\frac{1}{2}\omega\right) e^{\frac{1}{2}i\omega}$$

(II) Find the frequency response for $-1 \leq t \leq 2$.

Similarly, if we select $a = \frac{1}{2}$, delay the impulse response by 3 and a half seconds, and then scale by a factor of -1, we get the impulse response corresponding to $h(t) = \begin{cases} -1, \text{ if } 1 \leq t \leq 2 \\ 0, \text{ otherwise} \end{cases}$, thus giving that:

$$H_3(\omega) = -\,\text{sinc}\left(\frac{1}{2}\omega\right) e^{-\frac{3}{2}i\omega}$$

(III) Find the frequency response of the system.

Thus, putting $H_1(\omega)$ and $H_2(\omega)$ together, we find that:

$$\begin{aligned} H(\omega) &= H_1(\omega) + H_2(\omega) \\ &= \operatorname{sinc}\left(\frac{1}{2}\omega\right) e^{\frac{1}{2}i\omega} - \operatorname{sinc}\left(\frac{1}{2}\omega\right) e^{-\frac{3}{2}i\omega} \end{aligned}$$

Question 3

Answer the following questions:

(a) Compute the period of the signal given by $x(n) = \cos\left(\frac{5\pi}{4}n\right)$

Solution:

The period, P of $x(n)$ is the smallest value such that $\frac{5\pi}{4}P$ is a multiple of 2π where $P \in \mathbb{N}$. Thus, $P = 8$.

(b) Compute the Fourier series coefficients (X_k) of the signal given by $x(t) = 3 + 2\cos(t) + \sin(3t)$.

Solution:

By correlating the complex exponential expansion of each term in the signal, we arrive at the following Fourier series coefficients:

$$\begin{aligned} X_{-3} &= \frac{-1}{2i} & X_3 &= \frac{1}{2i} \\ X_{-2} &= 0 & X_2 &= 0 \\ X_{-1} &= 1 & X_1 &= 1 \\ & X_0 = 3 \end{aligned}$$

Note here that no signal has a frequency of 3, so $X_{-2} = X_2 = 0$.

(c) How fast must one sample to observe something occurring at a frequency of 10Hz?

Solution:

The sampling frequency must be strictly greater than twice the frequency of what we wish to observe. Thus, we must sample at a rate of > 20Hz.

(d) Compute the discrete Fourier transform of the signal $x(n) = \delta(n)$.

Solution:

The formula for the discrete Fourier transform is given by:

$$X_k = \sum_{n=0}^{N-1} x(n) e^{-i\omega_0 nk}$$

Therefore, we find that:

$$\begin{aligned}
X_k &= \sum_{n=0}^{N-1} x(n) e^{-i\omega_0 nk} \\
&= \sum_{n=0}^{N-1} \delta(n) e^{-i\omega_0 nk} \\
&= \delta(0) e^{-i\omega_0 0k} \\
&= e^0 \\
&= 1
\end{aligned}$$

Question 4

Given the signal $y(t) = x(t)\cos(\omega_c t)$. Given that the Fourier transform of $x(t)$ is $X(\omega)$, compute $Y(\omega)$.

Solution:

Note that multiplication in the time domain corresponds to convolution in the frequency domain. Thus, to find $y(t)$, we must find the frequency response of $\cos(\omega_c t)$ and convolute that with $X(\omega)$. Let $H(\omega)$ be the Fourier transform of $\cos(\omega_c t)$

(I) Compute $H(\omega)$.

To find $H(\omega)$, we must compute the continuous Fourier transform of $\cos(\omega_c t)$.

$$\begin{aligned} H(\omega) &= \int_{-\infty}^{\infty} \cos(\omega_c t)\, e^{-i\omega t} dt \\ &= \frac{1}{2}\int_{-\infty}^{\infty} \left(e^{i\omega_c t} + e^{-i\omega_c t}\right) e^{-i\omega t} dt \\ &= \frac{1}{2}\int_{-\infty}^{\infty} e^{it\omega(\omega_c - \omega)} + e^{-it(-\omega_c-\omega)} dt \end{aligned}$$

Now, using the fact that $\int_{-\infty} \infty e^{i\alpha t} dt = 2\pi\delta(a)$, we can conclude that:

$$\begin{aligned} H(\omega) &= \frac{1}{2}\left(2\pi\left(\delta(\omega_c - \omega) + \delta(\omega_c + \omega)\right)\right) \\ &= \pi\left(\delta(\omega_c - \omega) + \delta(\omega_c + \omega)\right) \end{aligned}$$

(II) Compute $Y(\omega)$.

As said earlier, since $y(t) = x(n)\cos(\omega_c t)$ and $X(\omega)$ and $H(\omega)$ are the frequency responses of $x(n)$ and $\cos(\omega_c t)$ respectively, we know that $Y(\omega) = X(\omega) * H(\omega)$. Thus:

$$\begin{aligned} Y(\omega) &= \int_{-\infty}^{\infty} X(\Omega) H(\omega - \Omega) d\Omega \\ &= \pi \int_{-\infty}^{\infty} X(\Omega)\left(\delta(\omega_c - (\omega - \Omega)) + \delta(\omega_c + (\omega - \Omega))\right) d\Omega \\ &= \pi \int_{-\infty}^{\infty} X(\Omega)\left(\delta(\omega_c - \omega + \Omega) + \delta(\omega_c + \omega - \Omega)\right) d\Omega \\ &= \pi\left(X(\omega - \omega_c) + X(\omega + \omega_c)\right) \end{aligned}$$

Question 5

(a) Compute the frequency response of a filter given by the difference equation:

$$\ddot{y} + 2\omega_0\dot{y} + {\omega_0}^2 y = x$$

Solution:

We know that the frequency response, $H(\omega)$, of the system is given by $y(t) = H(\omega)e^{i\omega t}$ when the input to the filter is given by $x(t) = e^{i\omega t}$. Thus, differentiating, we find that:

$$\dot{y}(t) = i\omega H(\omega)e^{i\omega t}$$

$$\begin{aligned}\ddot{y}(t) &= (i\omega)^2 H(\omega)e^{i\omega t} \\ &= -\omega^2 H(\omega)e^{i\omega t}\end{aligned}$$

Thus, by substituting, we find that:

$$\begin{gathered}\ddot{y} + 2\omega_0\dot{y} + {\omega_0}^2 y = x \implies \\ -\omega^2 H(\omega)e^{i\omega t} + 2\omega_0 i\omega H(\omega)e^{i\omega t} + {\omega_0}^2 H(\omega)e^{i\omega t} = e^{i\omega t} \\ -\omega^2 H(\omega) + 2\omega_0 i\omega H(\omega) + {\omega_0}^2 H(\omega) = 1 \\ H(\omega)\left(-\omega^2 + 2i\omega_0\omega + {\omega_0}^2\right) = 1 \\ H(\omega) = \frac{1}{-\omega^2 + 2i\omega_0\omega + {\omega_0}^2}\end{gathered}$$

(b) Design a minimum order, discrete filter that removes DC and passes the highest possible frequency with a gain of 1.

Solution:

(I) Construct a prototype filter.

Let $\tilde{H}(\omega)$ be the prototype filter which we will use to construct our final filter. We know that $\tilde{H}(0) = 0$ and $|H(\pi)| = 1$, since DC corresponds to a frequency of 0 and the highest possible discrete frequency is π. Thus, we find that:

$$\tilde{H}(\omega) = \left(1 - e^{-i\omega}\right)$$

Substituting in $\omega = \pi$:

$$\begin{aligned} |H(\pi)| &= |1 - e^{-i\pi}| \\ &= |1 - (-1)| \\ &= |2| \\ &= 2 \end{aligned}$$

(II) Construct the final filter.

Thus, normalizing our prototype filter by multiplying it by a factor of $\frac{1}{2}$, we get our final filter, $H(\omega)$, satisfying $|H(\pi)| = 1$. Thus:

$$H(\omega) = \frac{1}{2}\left(1 - e^{-i\omega}\right)$$

(III) Find the difference equation of the filter.

Multiplying both sides of our final filter by $e^{i\omega n}$, and recognizing that $y(n) = H(\omega)e^{i\omega n}$ when $e^{i\omega n}$, we find the difference equation of the system to be:

$$H(\omega) = \frac{1}{2}\left(1 - e^{-i\omega}\right) \implies$$

$$H(\omega)e^{i\omega n} = \frac{1}{2}\left(e^{i\omega n} - e^{-i\omega}e^{i\omega n}\right)$$

$$y(n) = \frac{1}{2}x(n) - \frac{1}{2}x(n-1)$$

Question 6

Compute the discrete Fourier transform of the signal $x(n) = \delta(n) - \delta(n-2)$ using $N = 4$.

Solution:

First, since $N = 4$, we can find that $\omega_0 = \frac{2\pi}{4} = \frac{\pi}{2}$
The discrete Fourier transform is given by the formula:

$$X_k = \sum_{n=0}^{N-1} x(n) e^{-i\omega_0 nk}$$

Thus, substituting in for $x(n)$ and N, we can simplify a little bit before computing the coefficients to help make things a little easier and less confusing.

$$\begin{aligned} X_k &= \sum_{n=0}^{N-1} x(n) e^{-i\omega_0 nk} \\ &= \sum_{n=0}^{3} \left(\delta(n) - \delta(n-2)\right) e^{-i\frac{\pi}{2}nk} \\ &= \sum_{n=0}^{3} \delta(n) e^{-i\frac{\pi}{2}nk} - \sum_{n=0}^{3} \delta(n-2) e^{-i\frac{\pi}{2}nk} \\ &= 1 - e^{-2i\frac{\pi}{2}k} \end{aligned}$$

Now, we can see that:

$$\begin{aligned} X_0 &= 1 - e^{-2i\frac{\pi}{2}(0)} = 1 - 1 = 0 \\ X_1 &= 1 - e^{-2i\frac{\pi}{2}(1)} = 1 - (-1) = 2 \\ X_2 &= 1 - e^{-2i\frac{\pi}{2}(2)} = 1 - 1 = 0 \\ X_3 &= 1 - e^{-2i\frac{\pi}{2}(3)} = 1 - (-1) = 2 \end{aligned}$$